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Comparative field test for measurement of PM_{10} dust in atmospheric air using gravimetric (reference) method and β-absorption method (Eberline FH 62-1)

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ABSTRACT

The paper presents the results of a field test carried out in Gdańsk region between 01-01-2010 and 31-12-2010 in order to demonstrate equivalence of the Eberline FH 62-1 sampler to the reference gravimetric method of suspended PM_{10} dust measurement. The differences in PM_{10} dust concentration provided by both methods have been discussed for different seasons of the year. A method of estimation of the correction factors/correction equations has been described based on the European Union guide "Demonstration of Equivalence of Ambient Air Monitoring Methods. EC Working Group on Guidance for the Demonstration of Equivalence". The results obtained from Eberline sampler are equivalent to the ones from gravimetric method, but after application of data correction. The investigation revealed different correction factors/correction equations for winter and summer season. The winter season is characterized by higher PM_{10} concentration and bigger discrepancies between both methods. A lack of equivalence has been found for PM_{10} dust concentration higher than 30 $\mu g\,m^{-3}$.

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1. Introduction

Particulate matter (PM) is a complex mixture of the molecules differing in chemical composition, shape and size. It can be emitted from both natural sources (erosion of rocks and soil, sea sprays, volcanic dust) as well as anthropogenic sources (fossil fuels combustion, traffic emission). A number of particles of a given size present in a particular place depends on a variety of factors including a source of PM, atmospheric processes: condensation, nucleation, evaporation, chemical reactions, settling on soil, leaching out due to precipitation. Some particles are emitted to the atmosphere directly from the sources, some are formed as a result of the aforementioned processes. Accordingly, PM_{10} is treated as the 'secondary' pollutant (Hinds, 1999). The chemical composition depends on the origin of dust and may include: simple anions (sulphates, nitrates, chlorides, silicates, carbonates), simple cations (ammonium, sodium, potassium, calcium, magnesium, aluminum ones), heavy metals (arsenic, cadmium, lead, nickel, chromium, vanadium) and adsorbed organic compounds such as polycyclic aromatic hydrocarbons (PAHs) (Rampazzo et al., 2008; van Drooge et al., 2009; Contini et al., 2010; Dongarra et al., 2010). There is a relatively strong correlation between suspended dust concentration in ambient air and PAHs content in this dust (Menichini, 2009). Numerous epidemiological investigations reveal statistical dependences between elevated dust concentration in air and negative health effects such as respiratory track and cardiovascular diseases (Viana et al., 2008; Schiliro et al., 2010). That is why continuous monitoring of suspended dust in air is of significant importance. The European Union established the requirements concerning admissible concentration of dust in ambient air (Directive of European Parliament and European Council 2008/50/WE of 21st May 2008) and obliged the member countries to undertake effective actions when the threshold concentration is exceeded.

The investigations show that PM_{10} (PM_{10} – all particles of diameter equal to 10 micrometers or smaller) dust concentration exceeds annual admissible level 40 $\mu g\,m^{-3}$ as well as 24-hour admissible level 50 $\mu g\,m^{-3}$ (24-hour admissible level can be exceeded 35 times during a year) in many European countries (Escudero et al., 2007; Gehrig et al., 2007; Vardoulakis and Kassomenos, 2008; Kleanthous et al., 2009; Burkart et al., 2010; Byrd et al., 2010). Provision of reliable information about dust concentration in air requires application of the dust meters operating according to a reference method or the method certified as equivalent to the reference method. The reference method for measurement of PM_{10} dust is a manual gravimetric method described in the EN 12341 standard. It provides dust concentration averaged over 24 hours, however the results are not available on-

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line (a filter with collected dust is stabilized in laboratory upon temperature and humidity conditions defined by the standard). However, there is a need for a measurement providing the concentration averaged over shorter time period (for instance 1 hour) available almost immediately.

Commonly utilized continuous samplers are: Eberline sampler (the measurement method based on beta radiation absorption) and TEOM sampler (the measurement method based on a shift of frequency of oscillatory balance) (Green et al., 2001; Viana et al., 2003). In order to show equivalence between the results obtained using the reference method and the continuous samplers some correction measures are taken consisting in application of the correction equation or correction factor.

The aim of this paper was evaluation if the automatic sampler Eberline FH 62-1 fulfilled the equivalence conditions. A comparison has been prepared for the entire year as well as for the winter and summer season separately. It has been verified if the same correction factors/equations could be used throughout the year or different ones should be employed in the particular seasons. Importance of the equivalent measurements in air quality monitoring employing the correction factors/equations cannot be underestimated. The potential erroneous measurements may take effect in two ways. On one hand, underrated results may endanger society to poor quality ambient air and bring health hazards. On the other hand, overrated results may launch unfounded correction actions/programmes yielding unjustified financial outlays.

Heterogeneous chemical composition of PM_{10} dust results in some difficulties in objective and reproducible evaluation of this pollutant content in air. Accuracy and stability of such measurements depend on many factors (Allen et al., 1997; Cyrys et al., 2001; Price et al., 2003), most often on two parameters: temperature and water vapour content. Changes of dust mass are caused by water vapour desorption, absorption of water vapour from air (hydration), chemical decomposition (volatilization) of the substances contained in dust, chemical reactions leading to formation of solid state substances (salts). These processes are dependent on temperature. In on-line samplers air is heated to constant, higher than ambient temperature. Such approach results in depletion of dust mass (ammonium salts, organic compounds) so the measurement results are lower than the ones obtained via the reference method. The difference depends on dust composition, measurement site, metrological parameters, season of the year and method of measurement (Speer et al., 2003; Charron et al., 2004; Green et al., 2009). An opposite situation concerns the reference method where a hysteresis phenomenon occurs during water desorption from quartz filters upon conditioning in the laboratory conditions of relative humidity $50 \pm 5\%$. Insufficient water desorption increases a mass of dust trapped in the filter. Water content in dust depends on dust composition (inorganic salts), relative humidity of air and it changes versus time as well as characteristics of local and regional pollution sources (Price et al., 2003; Speer et al., 2003). Generally, during winter season the on-line samplers underrate PM_{10} dust content as compared to the reference method, so a correction of the results is necessary. In summer season the differences between indications of the on-line sampler and the reference sampler are small, thus one can conclude an equivalence of the indications with a given probability level. Table 1 presents the correction factors for the Eberline samplers determined with respect to the reference sampler for the entire year in selected stations of the EU member countries disseminating information of air quality via the AirBase network (Buijsman and de Leeuw, 2004; de Leeuw, 2005; Larssen and de Leeuw, 2007). As it can be seen this type of sampler provides underrated values of PM_{10} dust concentration, the values of correction factors are generally within the range 1.0–1.3. However, in particular seasons reverse situation can occur that is the indications of the Eberline sampler are higher than the ones of the reference sampler. Such situations were observed in case of low concentrations of PM_{10} dust, low temperatures and high relative humidity of air (Chang et al., 2001; Salminen and Karlsson, 2003). Hauck et al. (2004a,b) observed that too long (one week) exposure of the quartz filters with trapped dust in the reference samplers in conditions of varying temperature resulted in a decrease in dust mass and in lower indications with respect to the Eberline sampler. Local changes of wind direction and velocity may also contribute to an increase in the indications of the Eberline sampler (Gehrig et al., 2009).

Table 1
Correction factors determined for the Eberline sampler with respect to the reference sampler for the entire year, obtained from selected measurement stations disseminating data via the AirBase network.

Country	Eberline sampler
Austria	1.18–1.3
Belgium	1.37
Estonia	1.15
Germany	1.0–1.3
Spain	0.84–1.3
Hungary	1.0–1.31
Lithuania	1.3
Netherlands	1.33
Portugal	1.11–1.18

2. Methodology of equivalence determination

A definition of the measurement method equivalent to the reference method anticipates that the results obtained using the candidate method will differ from the reference method results by the value not exceeding a statistically defined uncertainty. In case of PM_{10} dust measurements an extended uncertainty for the candidate methods (the Eberline sampler) fulfilling the equivalence conditions should be lower than 25%. Estimated measurement uncertainty is a sum of the uncertainty connected with random error and the uncertainty treated as a systematic error of the measurement methodology related to the admissible PM_{10} level (bias). According to the Directive of European Parliament and European Council 2008/50/WE one of the types of equivalence test is a field test. The test consists in placing both the reference and candidate samplers in a selected place and collecting information on PM_{10} dust concentration in air. A statistical tool, which makes it possible to determine the equivalence, is a non-standard guide proposed by the European Commission entitled "Demonstration of Equivalence of Ambient Air Monitoring Methods. EC Working Group on Guidance for the Demonstration of Equivalence". The guide recommends a minimum of 4 measurement campaigns consisting in 40 parallel measurements (in different seasons of the year) fulfilling the requirements of representativeness for an area characterized by a given level of PM_{10} dust concentration throughout a year. The criteria of acceptance of the candidate method as equivalent to the reference one in the field test are as follows:

- extended relative uncertainty U for parallel measurements lower than 25% (comparison of parallel measurements via orthogonal regression),
- inclination coefficient "b" not significantly different from unity, axis interception coefficient "a" (intercept) not significantly different from zero.

These criteria are applied to the entire measurement heat and to the measurement results above 30 $\mu g\, m^{-3}$ provided that $\geq$40 such

results were obtained. In case the acceptance criteria are not fulfilled a process of correction factor or correction equation evaluation is carried out. Lack of the correction factor or correction equation (extended uncertainty still above 25%) results in rejection of the candidate method and recognition of it as non-equivalent to the reference method.

Measurement uncertainty u_p obtained via orthogonal regression is determined from the equation (1):

$$u_p^2 = \frac{RSS}{n-2} - u_{RM}^2 + [a + (b-1)LV]^2 \quad (1)$$

where n is the number of parallel measurements, a is the intercept of the straight line obtained from regression, b is the inclination of the straight line obtained from regression, LV is the admissible level of dust concentration in air averaged over 24 hours (for PM_{10} dust – 50 $\mu g\,m^{-3}$), u_{RM}^2 is the measurement uncertainty between two reference samplers calculated from the dependence (2):

$$u_{RM}^2 = \frac{\sum_{i=1}^{n}(x_1 - x_2)^2}{2n} \quad (2)$$

where x_1 and x_2 are the results of parallel measurements of two reference samplers. A value of u_{RM} should fulfill the dependence (3) and in case of lack of information on u_{RM} an assumed value 0.82 ($u_{RM}^2 = 0.67$) should be applied in Eq. (1).

$$u_{RM} \le 2\ \mu g/m^3 \quad (3)$$

The parameter RSS present in Eq. (1) is a residual standard deviation calculated from the dependence (4):

$$RSS = \sum_{i=1}^{n}(y_{CM} - a - bx_{RM})^2 \quad (4)$$

where y_{CM} are the results obtained with the candidate method, x_{RM} denotes the results obtained with the reference method.

Relative uncertainty u_w is calculated from the dependence (5):

$$u_w = \frac{u_p}{LV} * 100\% \quad (5)$$

If the extended uncertainty U defined by Eq. (6) is at the level $\le$25% then the candidate method is equivalent to the reference method with 95% probability (provided the second condition of the acceptance criteria is fulfilled)

$$U = ku_w \quad (6)$$

where k is the coefficient of extension equal 2 for 95% certainty level. Else a correction process should be carried out. The correction factor or correction equation are calculated when the coefficients a and b of orthogonal linear regression are statistically significant or when the extended uncertainty is at the level $\ge$25%. Then three cases may occur:

- b is significantly different from unity, a is insignificantly different from zero, the correction consists in division of the results of candidate method y_{CM} by the value of coefficient b; then new orthogonal regression is applied to verify if extended uncertainty is at the level $\le$25% and if the coefficients of new regression are statistically insignificant (the first case),
- b is insignificantly different from unity, a is significantly different from zero, the correction consists in subtraction of the value of coefficient a from the results of candidate method y_{CM}; then new orthogonal regression is applied to verify if extended uncertainty is at the level $\le$25% and if the coefficients of new regression are statistically insignificant (the second case),
- b is significantly different from unity, a is significantly different from zero, the correction consists in subtraction of the value of coefficient a from the results of candidate method y_{CM}; then the obtained result is divided by the value of coefficient b; next new orthogonal regression is applied to verify if extended uncertainty is at the level $\le$25% and if the coefficients of new regression are statistically insignificant (the third case).

In case none of the aforementioned situations occurs the candidate method should be recognized as non-equivalent to the reference method.

3. Materials and methods

The measurements of PM_{10} dust concentration were performed from 01.01.2010 to 31.12.2010 at the measurement station AM8 of the ARMAAG Agency in Gdańsk (Agency of Regional Air Quality Monitoring in the Gdańsk Metropolitan Area ARMAAG). The station was equipped with a reference sampler HVS Digitel with a head for separation of PM_{10} dust and an automatic on-line analyzer Eberline FH 62-1. The aim of the measurements was a comparison of PM_{10} dust concentration obtained via gravimetric method (reference one) and automatic method. The other objectives were identification if the automatic method provides the results equivalent with the reference method and estimation of the correction factor/correction equation for the automatic method based on the aforementioned guide.

Mass of PM_{10} dust trapped on a quartz filter in HVS Digitel sampler was evaluated in an accredited laboratory, in a balance room, upon temperature and relative humidity specified in the PN-EN 12341 standard. The information of the volume of air filtered was obtained from a microprocessor module of the HVS Digitel sampler. The filters were supplied to the HVS Digitel sampler and back to the laboratory inside a dedicated transport container protecting them from mechanical damages and from a contact with the external factors. Volumetric flow rate of air through the samplers was as follows: 700 l min^{-1} for HVS, 16.7 l min^{-1} for Eberline. The volumes of air passed through all the devices were related to standard conditions that is to 20 °C and 1 atm. Calibration of the automatic sampler was carried out twice a year (flow and zero



Fig. 1. Location of urban background measurement station AM8 (ARMAAG) in Gdańsk.

Table 2
Results of parallel measurements of PM_{10} dust, averaging period – one year, obtained using the gravimetric method and the Eberline sampler. Extended measurement uncertainty for PM_{10} dust data before and after correction and estimation of the correction factor/correction equation.

Parameters	Reference method	Eberline sampler
Number of measurement days	365	
Number of significant days	353	
Average annual concentration ($\mu g\,m^{-3}$)	30.9	26.0
Number of days with the concentration higher than 50 $\mu g\,m^{-3}$	54	31
Maximum 24 h concentration ($\mu g\,m^{-3}$)	155.5	129.8
U – without data correction	37.6%	
U – with data correction	20.2%	
Correction factor	Lack	
Correction equation	$1.23C_{NR} - 1.07$	
Equivalence with the reference method	YES	

C_{NR} is the concentration of PM_{10} dust obtained from the Eberline sampler.

level), in case of the HVS sampler only flow was calibrated (the sampler was new and possessed calibration certificates). Additionally, the station was equipped with the temperature, relative humidity, wind direction and velocity sensors.

Urban background measurement station AM8 located in Gdańsk was selected for the field test (Fig. 1 presents localization of the station AM8). This station fulfils the representativeness requirements for an area characterized by a given level of PM_{10} dust concentration. It is placed beyond the impact of big, point emission sources. The site is mainly influenced by surface emission consisting of individual heating devices such as gas and oil heaters as well as domestic solid fuel furnaces. Traffic impact on this station is limited. The station is located in the area, where air quality standards were exceeded due to high concentrations of PM_{10} dust in recent years.

4. Results

4.1. Averaging period – one year

Table 2 presents the results of PM_{10} dust concentration measurements from the entire year obtained via the Eberline sampler and the gravimetric method. It can be noticed that in case of the gravimetric method the allowable annual frequency (35 times) of crossing the admissible dust concentration level was exceeded, what would result in different classification of air quality standards as compared to the automatic measurement method. A number of significant measurement data (measurement days) is determined based on a number of simultaneous measurements, during which both devices operated without disturbances (i.e. temporary failures, calibrations, gross errors). A statistical test confirms the equivalence of the measurements using the Eberline sampler with the reference method after data correction.

Extended measurement uncertainty evaluated for raw data (not subjected to correction) based on Eq. (6) was at the level 37.6% and orthogonal regression coefficients b and a were significantly different from 1 and 0, respectively. Data correction (see the third case) yielded extended measurement uncertainty at the level 20.2% and the coefficients of a new orthogonal regression were statistically insignificant from 1 and 0. It enabled estimation of the correction equation for the Eberline sampler, which is presented in Table 2. The correction of data from the Eberline sampler using the correction equation allowed identification of 50 measurement days, when PM_{10} dust concentration was above the admissible level (prior to the correction the number of such days was 31). The number of days characterized by concentration above the admissible level is comparable with the number obtained from the reference method (54).

Table 3
Results of parallel measurements of PM_{10} dust, averaging period – one year, where PM_{10} dust concentration exceeded 30 $\mu g\,m^{-3}$, obtained using the gravimetric method and the Eberline sampler. Extended measurement uncertainty for PM_{10} dust data before and after correction and estimation of the correction factor/correction equation.

Parameters	Reference method	Eberline sampler
Number of measurement days	120	
Number of significant days	119	
Average concentration ($\mu g\,m^{-3}$)	55.1	44.6
Number of days with the concentration higher than 50 $\mu g\,m^{-3}$	54	31
Maximum 24 h concentration ($\mu g\,m^{-3}$)	155.5	129.8
U – without data correction	44.4%	
U – with data correction	25.2%	
Correction factor	Lack	
Correction equation	Lack	
Equivalence with the reference method	NO	

Table 4
Results of parallel measurements of PM_{10} dust, averaging period – winter season, obtained using the gravimetric method and the Eberline sampler. Extended measurement uncertainty for PM_{10} dust data before and after correction and estimation of the correction factor/correction equation.

Parameters	Reference method	Eberline sampler
Number of measurement days	182	
Number of significant days	179	
Average concentration ($\mu g\,m^{-3}$)	41.6	33.2
Number of days with the concentration higher than 50 $\mu g\,m^{-3}$	52	30
Maximum 24 h concentration ($\mu g\,m^{-3}$)	155.5	129.8
U – without data correction	41.9%	
U – with data correction	19.2%	
Correction factor	Lack	
Correction equation	$1.19C_{NR} + 2.20$	
Equivalence with the reference method	YES	

C_{NR} is the concentration of PM_{10} dust obtained from the Eberline sampler.

4.2. Averaging period – one year, the measurement heat where PM_{10} dust concentration exceeded 30 $\mu g\,m^{-3}$

Table 3 gathers the results obtained using the Eberline sampler and the reference method from the entire year when PM_{10} dust concentration exceeded 30 $\mu g\,m^{-3}$. During a year period the number of days, when PM_{10} dust concentration in air was above

Table 5
Results of parallel measurements of PM_{10} dust, averaging period – summer season, obtained using the gravimetric method and the Eberline sampler. Extended measurement uncertainty for PM_{10} dust data before and after correction and estimation of the correction factor/correction equation.

Parameters	Reference method	Eberline sampler
Number of measurement days	183	
Number of significant days	174	
Average concentration ($\mu g\,m^{-3}$)	19.9	18.6
Number of days with the concentration higher than 50 $\mu g\,m^{-3}$	2	1
Maximum 24 h concentration ($\mu g\,m^{-3}$)	61.1	50.8
U – without data correction	21.2%	
U – with data correction	17.1%	
Correction factor	1.10	
Correction equation	Lack	
Equivalence with the reference method	YES	

Table 6
Results of parallel measurements of PM_{10} dust, averaging period – summer season, winter season obtained using the gravimetric method and the Eberline sampler in the period from 07-02-2007 to 09-02-2008 at the station AM9 (ARMAAG) Gdynia. Extended measurement uncertainty for PM_{10} dust data before and after correction and estimation of the correction factor/correction equation.

Parameters	Reference method	Eberline sampler
Average concentration in summer ($\mu g\,m^{-3}$)	16.6	15.6
U – without data correction	14.3%	
U – with data correction	14.3%	
Correction factor	Lack	
Correction equation	Lack	
Equivalence with the reference method	YES	
Average concentration in heating season ($\mu g\,m^{-3}$)	27.3	22.8
U – without data correction	40.0%	
U – with data correction	17.0%	
Correction factor	Lack	
Correction equation	1.28 C_{NR}–2.00	
Equivalence with the reference method	YES	

C_{NR} is the concentration of PM_{10} dust obtained from the Eberline sampler.

30 $\mu g\,m^{-3}$ is equal to 119. 100 out of 119 days belong to the winter season (heating season). Extended measurement uncertainty for raw data was at the level 44.4%, whereas data correction (see the third case) led to extended measurement uncertainty equal to 25.2%. This level of the uncertainty suggests lack of equivalence of the Eberline sampler. However, extended uncertainty was close to the limit (25%) and in this case inferring about non-equivalence of the Eberline sampler can lead to erroneous conclusion. In our opinion the main reason is the fact that almost all results concern the winter season (December, January, February) when the lowest temperature and the highest relative humidity occurred (see Table 7).

In Poland heating season (heating in individual households) is from October till March, but October, November and March are relatively warm months due to the greenhouse effect. In effect these three coldest months generate the highest PM_{10} dust concentration and a difference in concentration level obtained via both methods is the highest. As a result there could be both underestimation and overestimation effect. The underestimation of the Eberline sampler can be due to a loss of volatile compounds trapped on the surface of dust suspended in air. The overestimation can originate from a phenomenon of humidity hysteresis on quartz filters can occur in the reference sampler. It leads to exaggerated concentration of PM_{10} dust in this measurement period (Price et al., 2003). Unfortunately, this effect has not been investigated by us in details, thus no unequivocal conclusions can be stated in this case, when extended uncertainty at the criterion limit may lead to mistakes.

4.3. Averaging period – winter (heating) season

Table 4 presents the results obtained using the Eberline sampler and the reference method for the period from 01-01-2010 to 30-03-2010 and from 01-10-2010 to 31-12-2010.

Extended measurement uncertainty for raw data was at the level 41.9%. Data correction (see the third case) provided extended measurement uncertainty at the level 19.2% and the coefficients of a new orthogonal regression were statistically insignificant from 1 and 0. It enabled estimation of the correction equation for the Eberline sampler, which is given in Table 4. The correction of data from the Eberline sampler using the correction equation allowed identification of 49 measurement days when PM_{10} dust concentration was above the admissible level (prior to the correction the number of such days was 30). The number of days with concentration above the admissible level is similar to the number obtained from the reference method (52). Additional 79 measurement days in the heating season, as compared to Section 4.2, provided certain conclusion about equivalence of the Eberline sampler. Analyzing data from Tables 3, 4 and 7 one can clearly notice an impact of temperature and relative humidity on the results obtained from the reference method and from the Eberline sampler.

4.4. Averaging period – summer season

Table 5 presents the results obtained using the Eberline sampler and the reference method for the period from 01-04-2010 to 30-09-2010. Extended measurement uncertainty estimated for raw data was at the level 21.2%. Although the first criterion was fulfilled $U \leq 25\%$, the second one was not – the coefficient b of orthogonal regression was statistically significant from 1. Accordingly, data correction should have been carried out following the first case. After data correction extended measurement uncertainty amounted 17.1% and the coefficients of a new orthogonal regression were statistically insignificant from 1 and 0. It made possible to evaluate the correction factor for the Eberline sampler included in Table 5. Comparing Tables 4 and 5 (winter and summer season) one can observe that in the winter season PM_{10} dust concentration as well as the differences in the results obtained using both methods are higher.

Comparing an analogous field test performed between 07-02-2007 and 09-02-2008 on the AM9 station (Table 6) located in Gdynia in Tri-City region (Gębicki and Szymańska, 2011) convergent information can be found. During the summer season average PM_{10} dust concentration obtained using the reference method and the Eberline sampler was equal to 16.6 and 15.6 $\mu g\,m^{-3}$, respectively, while extended measurement uncertainty was at the level 14.3% and there was no need to apply the correction factor/equation. In the winter season the corresponding average PM_{10}

Table 7
Monthly averaged values of PM_{10} dust concentration, temperature, relative humidity, wind velocity and direction.

Month	Concentration reference method [$\mu g\,m^{-3}$]	Concentration Eberline sampler [$\mu g\,m^{-3}$]	Temperature [°C]	Relative humidity [%]	Wind velocity [$m\,s^{-1}$]	Wind direction [°]
January	53.2	45.3	−6.3	84.0	2.1	159
February	56.0	44.7	−1.4	83.3	1.9	159
March	34.3	26.8	3.7	78.2	2.3	225
April	29.5	23.6	7.6	72.2	2.2	235
May	19.4	15.7	11.4	78.4	1.8	216
June	19.2	17.2	16.6	69.4	1.6	232
July	19.0	19.0	21.8	70.0	1.6	208
August	16.4	18.3	19.8	76.3	1.3	201
September	15.8	17.6	13.8	78.1	1.8	184
October	29.7	25.8	7.1	76.7	1.9	199
November	27.9	19.8	5.2	85.6	2.0	184
December	48.8	37.0	−5.0	85.3	2.2	200

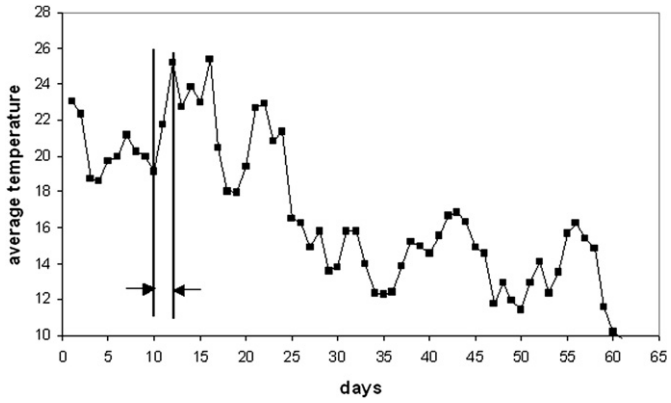


Fig. 2. Changing the air temperature in the months of August and September at the measuring station AM8 (ARMAAG) in Gdańsk.

concentrations were 27.3 and 22.8 $\mu g\,m^{-3}$. Extended measurement uncertainty was at the level 40.0% and after data correction (the third case) it was equal to 17.0%. Data correction allowed determination of the correction equation (presented in Table 6) for the Eberline sampler in the winter season. All correction factors/equations estimated during particular measurement periods for both AM8 and AM9 stations were at the level presented in Table 1.

4.5. Averaging period – one month

Table 7 gathers monthly averaged PM_{10} dust concentration, temperature, relative humidity, wind velocity and direction. In case of summer season it can be observed that for two months August–September average PM_{10} dust concentration obtained using the Eberline sampler was higher than the corresponding concentration from the reference method. These were the two months in the entire year when average concentration obtained from the gravimetric method was the lowest. Figs. 2 and 3 reveal that during short time intervals (a few days) there was a rapid air temperature change (Fig. 2) reaching 9 °C and a change in relative humidity up to 30% (Fig. 3). Figs. 2 and 3 present only exemplary increases and decreases of meteorological data. Such rapid changes of meteorological data during short time have negative influence on the quartz filters containing trapped dust, which are placed inside the reference samplers. Upon varying temperature and relative humidity dust mass decrement occurs resulting in lower concentrations as compared to the Eberline sampler.

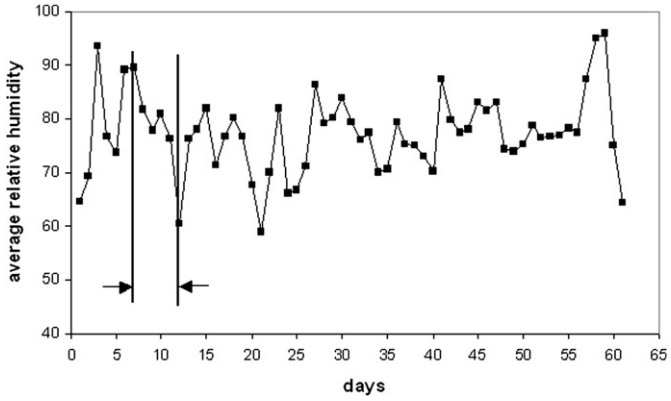


Fig. 3. Changing the relative humidity in the months of August and September at the measuring station AM8 (ARMAAG) in Gdańsk.

During remaining months average PM_{10} dust concentration provided by the gravimetric method was higher than the corresponding one from the Eberline sampler. During the months, when temperature was below 0°C, average difference in PM_{10} dust concentration obtained via both samplers was the highest.

5. Conclusions

The field test for PM_{10} dust concentration, performed from 01.01.2010 to 31.12.2010 at the measurement station AM8 of the ARMAAG Agency in Gdańsk, revealed that the results obtained from the Eberline sampler were equivalent to the ones provided by the gravimetric method after data correction. The investigations showed that there was not a single correction factor or a single correction equation for the entire time of field test duration. During winter season PM_{10} dust concentration as well as the differences in the results obtained using both methods are higher. Moreover, for the dust concentration higher than 30 $\mu g\,m^{-3}$ the equivalence conditions are not fulfilled. It confirms a recognized regularity (Speer et al., 2003; Charron et al., 2004; Green et al., 2009) that in winter season, when average PM_{10} dust concentration is the highest during the year, the Eberline sampler results are underrated as compared to the gravimetric method. A loss of volatile components, high relative humidity, low temperature have a negative impact on the processes of sorption-desorption of water vapour on the quartz filter causing an overrating of PM_{10} dust concentration provided by the gravimetric method (Price et al., 2003). The analysis of averaged measurement periods (entire year, winter season, summer season) revealed that the evaluated correction factors/correction equations fell within the range described in Table 1. The situation during August-September period, when average PM_{10} dust concentration obtained using the Eberline sampler was higher than the corresponding concentration provided by the reference method, confirms the observations reported in Hauck et al. (2004a,b). The field test results show that average extended uncertainty of the results from the Eberline sampler determined with the respect to the gravimetric method was at the level of 20% (after data correction). Such level of extended uncertainty makes one to apply and obey due diligence (due care ensuring the quality of measurements) through: comparative campaigns of candidate samplers (in order to determine uncertainty u^2_{RM}), measurement campaigns at different measurement sites, measurement quality control (flow calibration, calibration of automatic samplers, intercalibration, head maintenance).

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